

DSBDA ORAL EXAM PREPARATION GUIDE

Data Science and Big Data Analytics Laboratory (310256)

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ASSIGNMENT 1: Data Wrangling I | Dataset: Iris / Any open-source CSV (Kaggle)

Main Logic of the Program

■ Load any CSV dataset → check missing values using `isnull()` → check data types with `dtypes` → apply Min-Max Normalization → convert categorical columns (Species) using Label Encoding / One-Hot Encoding.

Q1. What is Data Wrangling?

Ans: Data Wrangling (also called Data Munging) is the process of cleaning, transforming, and organizing raw data into a usable format for analysis. It includes handling missing values, fixing data types, normalization, and encoding.

Q2. What is a Pandas DataFrame?

Ans: A DataFrame is a 2D table-like data structure in Pandas with labeled rows and columns. It can hold different data types per column. Example: `iris = pd.read_csv('iris.csv')` loads data as a DataFrame.

Q3. How do you check and handle missing values?

Ans: Use `df.isnull()` to get True/False for each cell. `df.isnull().sum()` gives count per column. Handle by: `df.fillna(df.mean())` — replace with mean, or `df.dropna()` — remove rows with missing values.

Q4. What is Data Normalization and why is it needed?

Ans: Normalization scales all numeric values to a common range (0 to 1) using Min-Max scaling: $(x - \min) / (\max - \min)$. It is needed so that larger-valued features don't dominate smaller-valued ones during model training.

Q5. What is the difference between Label Encoding, One-Hot Encoding, and Dummy Encoding?

Feature	Label Encoding	One-Hot Encoding	Dummy Encoding
Number of columns	Same (1 column)	k columns for k classes	k-1 columns
Creates priority issue?	YES	No	No
Best used when	2 classes only	Multi-class, no order	Avoids multicollinearity
Function used	<code>LabelEncoder()</code>	<code>OneHotEncoder()</code>	<code>pd.get_dummies()</code>

Q6. What is the limitation of Label Encoding?

Ans: Label Encoding assigns numbers 0, 1, 2... to categories. This creates a false priority: the model may treat category '2' as greater than '0', which is wrong for unordered categories like flower species.

Q7. Name important Pandas functions used in Data Preprocessing.

Ans: `df.head()`, `df.tail()`, `df.shape`, `df.dtypes`, `df.describe()`, `df.isnull().sum()`, `df.dropna()`, `df.fillna()`, `df.astype()`, `df.sort_values()`, `df.iloc[]`, `df.loc[]`

■ *Tip: If asked 'Which dataset?' say: Iris dataset from UCI repository (150 rows, 5 columns: SepalLength, SepalWidth, PetalLength, PetalWidth, Species).*

ASSIGNMENT 2: Data Wrangling II | Dataset: StudentsPerformance (Custom CSV, 30 rows)

Main Logic of the Program

■ Create a custom student dataset in Excel → load it → scan for null values using `isnull()` → detect outliers using Boxplot / Z-Score / IQR → remove or replace outliers → apply data transformation (log scale or feature construction).

Q1. What is an Outlier? Why does it matter?

Ans: An outlier is a data point that is very different from the rest (too high or too low). Example: In [15, 16, 18, 101, 13], '101' is an outlier. Outliers affect the Mean most — pulling it away from the real center of data.

Q2. What are the methods to detect outliers?

Ans: 1) Boxplot: Visual method using Q1, Q2, Q3. 2) Scatter Plot: Checks relationship between two variables. 3) Z-Score: $Z = (x - \text{mean}) / \text{std_dev}$. Points with $|Z| > \text{threshold}$ are outliers. 4) IQR Method: $\text{IQR} = Q3 - Q1$. Outlier if $x < Q1 - 1.5 * \text{IQR}$ or $x > Q3 + 1.5 * \text{IQR}$.

Q3. What are the methods to handle outliers?

Ans: 1) Trimming: Remove the outlier row from dataset using `df.drop()`. 2) Capping (Quantile-based): Replace values above 90th percentile with 90th percentile value. 3) Mean/Median Imputation: Replace outlier with median of that column (preferred, since median is not affected by outliers).

Q4. What is Data Transformation and why is it needed?

Ans: Data transformation converts raw data into a better format. It is needed to: change scale for better understanding (e.g., joining year → duration), or reduce skewness in data by applying log transformation (`np.log10`).

Q5. What is the difference between `isnull()` and `notnull()`?

Ans: `isnull()` returns True where value is missing (NaN). `notnull()` returns True where value is present (not NaN). Both return a Boolean DataFrame.

Q6. What is skewness? How do you reduce it?

Ans: Skewness is asymmetry in data distribution. Positive skew: tail on right (mean > median). Negative skew: tail on left (mean < median). To reduce: apply log transformation using `np.log10(column)`, which compresses large values and brings data closer to normal distribution.

■ *Tip: Dataset used: StudentsPerformance.csv — custom created in Excel with 30 instances, features: Math_Score, Reading_Score, Writing_Score, Placement_Score, Club_Join_Date, Placement_Offer_Count.*

ASSIGNMENT 3: Descriptive Statistics | Dataset: Iris.csv + Mall Customers / Any dataset

Main Logic of the Program

■ Load dataset → compute mean/median/mode/std/min/max using `df.describe()` → group by categorical variable using `groupby()` → display statistical details per species of Iris using `describe()`.

Q1. What are Measures of Central Tendency?

Ans: They describe the center/middle of data. Three types: 1) Mean: Sum of all values / total count (affected by outliers). 2) Median: Middle value when data is sorted (not affected by outliers). 3) Mode: Most frequently occurring value. Data can be uni-modal, bi-modal, or multi-modal.

Q2. What are Measures of Dispersion?

Ans: They describe how spread out the data is. Include: Range (max - min), Variance (avg of squared differences from mean), Standard Deviation (square root of variance), Quartiles (Q1=25%, Q2=50%, Q3=75%), and Skewness (asymmetry).

Q3. What is the difference between Descriptive and Inferential Statistics?

Ans: Descriptive Statistics: Summarizes the available data (mean, median, std) — no generalizing. Inferential Statistics: Uses a sample to make predictions/inferences about the whole population using probability theory.

Q4. Which Python functions are used to compute statistics?

Ans: `df.mean()`, `df.median()`, `df.mode()`, `df.std()`, `df.min()`, `df.max()`, `df.describe()` — gives all stats at once. `df.groupby(['Category'])['Value'].mean()` — grouped statistics.

Q5. What are the types of variables?

Ans: Categorical: Nominal (no order, e.g., color, city) and Ordinal (with order, e.g., rating: Good > Average > Poor). Numerical: Continuous (infinite values in a range, e.g., height) and Discrete (finite values, e.g., number of students).

Q6. How does mean differ from median when outliers are present?

Ans: Mean gets pulled toward outliers (higher or lower). Median stays near the center and is unaffected. So, when data has outliers, Median is a better measure of central tendency.

ASSIGNMENT 4: Data Analytics I — Linear Regression | Dataset: Boston Housing Dataset

Main Logic of the Program

■ Load Boston dataset → split into X (13 features) and Y (PRICE) → split into 80% train / 20% test using `train_test_split()` → fit LinearRegression model → predict on test set → evaluate using MSE and R-Squared.

Q1. What is Linear Regression?

Ans: Linear Regression is a supervised machine learning algorithm that finds a straight-line relationship between input (X) and output (Y). Formula: $Y = mX + b$, where m = slope, b = intercept. It predicts continuous output values.

Q2. What is the difference between Simple and Multiple (Multivariate) Linear Regression?

Ans: Simple Linear Regression: One input feature (X) predicts one output (Y). Equation: $Y = a + bX$. Multiple Linear Regression: Two or more features predict one output. Equation: $Y = a + b_1X_1 + b_2X_2 + \dots$. Boston dataset uses multiple regression (13 features).

Q3. What dataset is used and what is its structure?

Ans: Boston Housing Dataset: 506 samples, 14 features (e.g., CRIM=crime rate, RM=avg rooms, LSTAT=lower status), and target = PRICE (median house value in \$1000s). From `sklearn.datasets` or Kaggle.

Q4. What is MSE and R-Squared? How do you interpret them?

Ans: MSE (Mean Squared Error): Average of squared differences between actual and predicted values. Lower MSE = better model. R-Squared: Measures how much variance the model explains. Value between 0 and 1. Closer to 1 = better fit (e.g., $R^2=0.48$ means model explains 48% of variance).

Q5. What is the difference between Training error and Testing error?

Ans: Training Error (E_{in}): Error on data the model was trained on — usually low. Testing Error (E_{out}): Error on new unseen data — generally higher than training error. A large gap means the model is overfitting.

Q6. What is the Least Squares Method?

Ans: It finds the best-fit line by minimizing the sum of squared differences between actual and predicted values. The slope (β_1) = $\frac{\sum[(x-x_{\text{mean}})(y-y_{\text{mean}})]}{\sum[(x-x_{\text{mean}})^2]}$ and intercept (β_0) = $y_{\text{mean}} - \beta_1 * x_{\text{mean}}$.

■ *Tip: Key functions: `LinearRegression().fit()`, `lm.predict()`, `train_test_split()`, `r2_score()`, `mean_squared_error()`*

ASSIGNMENT 5: Data Analytics II — Logistic Regression | Dataset: Social_Network_Ads.csv

Main Logic of the Program

■ Load `Social_Network_Ads.csv` → select features (Age, Salary) and target (Purchased: 0 or 1) → split train/test → fit LogisticRegression model → predict → compute Confusion Matrix → calculate Accuracy, Precision, Recall, F1-Score.

Q1. What is Logistic Regression? How is it different from Linear Regression?

Feature	Linear Regression	Logistic Regression
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Output type	Continuous (e.g., price)	Categorical (0 or 1)
Used for	Regression problems	Classification problems
Estimation method	Ordinary Least Squares (OLS)	Maximum Likelihood (MLE)
Output function	Straight line	Sigmoid/S-curve (0 to 1)
Example	House price prediction	Email spam or not spam

Q2. What is the Sigmoid Function?

Ans: The Sigmoid function maps any real number to a value between 0 and 1. It gives an S-shaped curve. If output > 0.5 → predict class 1 (Yes/Positive). If output < 0.5 → predict class 0 (No/Negative). Formula: $1 / (1 + e^{(-x)})$

Q3. What is a Confusion Matrix? Explain TP, TN, FP, FN.

Ans: A Confusion Matrix shows actual vs predicted values in a table: TP (True Positive): Predicted YES, Actual YES. TN (True Negative): Predicted NO, Actual NO. FP (False Positive): Predicted YES, but Actual NO (Type I error). FN (False Negative): Predicted NO, but Actual YES (Type II error).

Q4. What are Accuracy, Precision, Recall, and F1-Score?

Ans: Accuracy = $(TP+TN)/(TP+TN+FP+FN)$ — overall correctness. Precision = $TP/(TP+FP)$ — out of all predicted positives, how many are correct. Recall = $TP/(TP+FN)$ — out of all actual positives, how many did we catch. F1-Score = $2*Precision*Recall/(Precision+Recall)$ — harmonic mean of both.

Q5. What are the types of Logistic Regression?

Ans: Binary: 2 output classes (e.g., spam or not). Multinomial: 3+ unordered classes (e.g., type of wine). Ordinal: 3+ ordered classes (e.g., rating 1 to 5).

■ *Tip: Social_Network_Ads.csv: Columns = User ID, Gender, Age, Estimated Salary, Purchased (0/1). Features used: Age & Salary. Target: Purchased.*

ASSIGNMENT 6: Data Analytics III — Naive Bayes | Dataset: Iris.csv

Main Logic of the Program

■ Load Iris dataset → encode Species (0,1,2) → split into X (4 features) and Y (Species) → train/test split → apply GaussianNB() → predict → compute Confusion Matrix → calculate Accuracy, Precision, Recall.

Q1. What is Naive Bayes? Why is it called 'Naive'?

Ans: Naive Bayes is a classification algorithm based on Bayes' Theorem. It is called 'Naive' because it assumes all features are independent of each other — which is a simplification (rarely true in real data), but it still works well in practice.

Q2. What is Bayes' Theorem? State it simply.

Ans: $P(\text{Class} | \text{Features}) = P(\text{Features} | \text{Class}) * P(\text{Class}) / P(\text{Features})$. In simple words: the probability that a data point belongs to a class, given its features, is proportional to the probability of seeing those features in that class multiplied by how common that class is.

Q3. What is Prior Probability and Conditional Probability?

Ans: Prior Probability: Probability calculated with no other information (e.g., $P(\text{Yes}) = 9/14$ in a dataset). Conditional Probability: Probability of one event given another has occurred (e.g., $P(\text{Sunny} | \text{Yes}) =$ probability of sunny weather given we play).

Q4. Difference between Naive Bayes and Logistic Regression?

Ans: Naive Bayes: Probabilistic, based on Bayes Theorem, works well with small data, assumes feature independence. Logistic Regression: Discriminative model, based on Sigmoid function, works better with large data, no independence assumption. Both are classification algorithms.

Q5. What is GaussianNB and when is it used?

Ans: GaussianNB is a variant of Naive Bayes used when features are continuous (numeric). It assumes features follow a Gaussian (normal) distribution. Used for Iris dataset since features like SepalLength,

PetalWidth are continuous numbers.

■ *Tip: Iris.csv: 150 rows, 3 species (Setosa, Versicolor, Virginica), 4 numeric features. GaussianNB from sklearn.naive_bayes is used.*

ASSIGNMENT 7: Text Analytics | Dataset: Sample text / Kaggle Movie Reviews / Tweets

Main Logic of the Program

■ Take a sample text document → apply Tokenization (break into words/sentences) → remove Stop Words → apply Stemming (chop suffix) → apply Lemmatization (get base word using dictionary) → apply POS Tagging → calculate TF-IDF values.

Q1. What is Tokenization?

Ans: Tokenization is the process of breaking a text paragraph into smaller units called tokens — either words (word tokenization) or sentences (sentence tokenization). It is the first step in NLP processing. Library: NLTK — `sent_tokenize()`, `word_tokenize()`.

Q2. What are Stop Words and why are they removed?

Ans: Stop words are common words that appear frequently but carry little meaning — like 'is', 'the', 'a', 'and', 'are'. They are removed to reduce noise and focus the model on meaningful words. NLTK provides a built-in list: `from nltk.corpus import stopwords`.

Q3. What is the difference between Stemming and Lemmatization?

Feature	Stemming	Lemmatization
Method	Chops off suffixes (crude)	Uses dictionary & grammar (smart)
Result	May not be a real word	Always a real word (lemma)
Example	running → run, better → bett	running → run, better → good
Speed	Faster	Slower
Tool	PorterStemmer	WordNetLemmatizer

Q4. What is POS Tagging?

Ans: Part-of-Speech (POS) Tagging identifies the grammatical role of each word in a sentence — Noun (NN), Verb (VB), Adjective (JJ), Pronoun (PRP), Adverb (RB), etc. Example: 'The dog runs fast' → [The/DT, dog/NN, runs/VBZ, fast/RB].

Q5. What is TF-IDF? Explain simply.

Ans: TF-IDF (Term Frequency - Inverse Document Frequency) measures how important a word is in a document compared to all documents. TF = how often word appears in a doc. IDF = $\log(\text{total docs} / \text{docs containing word})$. $\text{TF-IDF} = \text{TF} * \text{IDF}$. High TF-IDF means word is unique and important to that document.

Q6. What is the Bag of Words (BoW) model?

Ans: BoW represents text as a matrix counting how many times each word appears in each document, ignoring order and grammar. It converts text into numbers that a machine learning model can use. Simple but loses word order and context.

ASSIGNMENTS 8, 9, 10: Data Visualization I, II, III | Datasets: Titanic (8,9), Iris (10)

Main Logic of the Program

■ Assign 8: Load Titanic → plot Histogram of 'fare' column using `sns.histplot()`. Assign 9: Plot Box plot of 'age' grouped by 'sex' and 'survived' using `sns.boxplot()`. Assign 10: Load Iris → list features → plot histogram and boxplot for each feature → identify outliers.

Q1. What is the difference between Matplotlib and Seaborn?

Feature	Matplotlib	Seaborn
Built on	Base library	Built on top of Matplotlib
Best for	Basic plots (line, bar)	Statistical visualizations
Works with	Arrays and datasets	Entire datasets (DataFrames)
Code complexity	More code needed	Complex plots with less code
Customization	Highly customizable	Pre-built themes

Q2. What is a Histogram? When do you use it?

Ans: A Histogram shows the distribution of a single numerical variable by grouping values into ranges (bins) and showing the count/frequency of each bin as a bar. Use when you want to see how data is distributed (e.g., how ticket fares are spread in Titanic).

Q3. What is a Box Plot? What does it show?

Ans: A Box Plot (Whisker plot) shows the 5-number summary: Minimum, Q1 (25%), Median (50%), Q3 (75%), Maximum. The box represents Q1 to Q3, line in box = median, whiskers extend to min/max. Points beyond whiskers are outliers. Used to compare distributions across groups.

Q4. What patterns can you observe from Titanic dataset visualizations?

Ans: Histogram of fare: Most passengers paid low fares (right-skewed distribution). Box plot of age vs sex vs survived: Female survivors tend to be of varied ages; young children had higher survival. First class passengers had higher fares and better survival rates.

Q5. What are the features of the Iris dataset used in Assignment 10?

Ans: 4 Numeric features: SepalLengthCm, SepalWidthCm, PetalLengthCm, PetalWidthCm. 1 Categorical: Species (Iris-setosa, Iris-versicolor, Iris-virginica). Total: 150 rows, 3 species with 50 samples each.

Q6. How does a box plot help identify outliers?

Ans: Points that fall beyond the whiskers (above $Q3 + 1.5 \cdot IQR$ or below $Q1 - 1.5 \cdot IQR$) are shown as individual dots in a box plot — these are outliers. This gives a quick visual way to spot unusual data points.

GROUP B: Big Data — Hadoop MapReduce & Scala | Input: text files / log files

Main Logic

■ B1 (Word Count): Map function tokenizes text and emits (word, 1) pairs. Reduce function sums counts for each word. B2 (Log file): Map splits log by '-' and emits (country, 1). Reduce sums. B3 (Scala/Spark): Simple Scala programs run in IntelliJ or spark-shell.

Q1. What is Hadoop? What problem does it solve?

Ans: Hadoop is an open-source framework for storing and processing very large datasets (Big Data) across a cluster of computers. It solves the problem of handling data too large for a single machine by distributing storage (HDFS) and processing (MapReduce) across many nodes.

Q2. Explain MapReduce with the Word Count example.

Ans: MapReduce has two phases: Map Phase: Takes input text, splits into lines, tokenizes words, emits key-value pairs: (Pune,1), (Mumbai,1), (Pune,1). Reduce Phase: Groups by key and sums: (Mumbai,1), (Nashik,2), (Pune,2). The Combiner is an optional local reducer that runs before sending to Reducer.

Q3. What are the components of Hadoop Ecosystem?

Ans: HDFS (storage), YARN (resource manager), MapReduce (processing), Pig (scripting), Hive (SQL-like), HBase (NoSQL database), Spark (fast in-memory processing), Mahout (machine learning), Solr/Lucene (search/indexing).

Q4. What is HDFS? What are NameNode and DataNode?

Ans: HDFS (Hadoop Distributed File System) stores large files across multiple machines. NameNode: Master node — stores metadata (file names, locations, directory structure). Does NOT store actual data. DataNode: Stores actual data blocks. Multiple DataNodes = fault tolerance via replication (usually 3 copies).

Q5. What is YARN?

Ans: YARN (Yet Another Resource Negotiator) is the resource management layer of Hadoop. It has two parts: ResourceManager (master — allocates resources, schedules tasks) and NodeManager (worker — runs tasks on each DataNode, monitors resource usage).

Q6. What is Apache Spark? How is it different from MapReduce?

Ans: Spark is a fast, in-memory distributed data processing framework. It is 100x faster than MapReduce because it stores intermediate results in RAM instead of writing to disk. MapReduce reads/writes disk at every step (slow). Spark supports Python, Scala, Java, R.

Q7. What is Apache Hive? How is it different from Pig?

Ans: Hive: SQL-like interface for Hadoop (HQL = HiveQL). Best for people who know SQL. Used for batch query processing. Pig: Uses Pig Latin scripting language. Best for ETL (Extract, Transform, Load). 10 lines of Pig Latin = ~200 lines of MapReduce Java code. Both convert to MapReduce internally.

IMPORTANT GENERAL VIVA QUESTIONS

Q1. What is Data Science?

Ans: Data Science is the field of extracting meaningful insights from structured and unstructured data using statistics, programming, and machine learning. It involves data collection, cleaning, analysis, visualization, and modeling.

Q2. What is the difference between Supervised and Unsupervised Learning?

Ans: Supervised Learning: Model is trained on labeled data (input + output given). Examples: Linear Regression, Logistic Regression, Naive Bayes. Unsupervised Learning: Model finds patterns in unlabeled data. Examples: K-Means Clustering, PCA.

Q3. What Python libraries are used in DSBDA practicals?

Ans: Pandas (data manipulation), NumPy (arrays), Matplotlib (basic plotting), Seaborn (statistical visualization), Scikit-learn (ML algorithms), NLTK (NLP/text processing).

Q4. What is train_test_split and why is it used?

Ans: train_test_split() from sklearn divides the dataset into training set (to train the model) and test set (to evaluate performance on unseen data). Common split: 80% train, 20% test. This prevents overfitting and gives a fair performance estimate.

Q5. What is Overfitting and Underfitting?

Ans: Overfitting: Model learns training data too well (including noise), performs poorly on new data (high training accuracy, low test accuracy). Underfitting: Model is too simple, performs poorly even on training data. Solution: Use proper train/test split, regularization, or more/less complex models.

Q6. What is the difference between Classification and Regression?

Ans: Classification: Predicts a category/class (discrete output). Examples: spam/not spam, species type. Algorithms: Logistic Regression, Naive Bayes. Regression: Predicts a continuous numeric value. Examples: house price, salary. Algorithm: Linear Regression.

Q7. What is Big Data? Explain the 3 Vs.

Ans: Big Data refers to datasets too large/complex for traditional tools to handle. 3 Vs: Volume (huge amount of data — terabytes/petabytes), Velocity (data generated at high speed — real-time streaming), Variety (different types — structured, unstructured, semi-structured).